Global Nitrogen Fertilizer Use Intensity, 1961–2023: Trends and Economic Implications for Agricultural Nutrient Pollution

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I. Introduction

This paper studies the agricultural use of inorganic nitrogen fertilizer per hectare of cropland. The variable is measured in kilograms of nitrogen nutrient per hectare, or kg N/ha. Its numerator is the nitrogen contained in inorganic fertilizer used in agriculture. Its denominator is cropland, defined as arable land plus permanent crops (Food and Agriculture Organization of the United Nations [FAO], 2025). The measure shows how intensively farmers apply synthetic nitrogen to cropland. It does not measure nitrate in drinking water or the amount of nitrogen that leaves a field.

Nitrogen fertilizer creates both benefits and costs. Crops need nitrogen, and fertilizer can raise yields when nitrogen is limited. Some nitrogen is not taken up by crops, however. It can reach water through runoff and leaching, contribute to eutrophication, or leave the field through other pathways. Farmers receive the value of the additional crop, while part of the environmental cost can fall on water users, utilities, fisheries, and the public (European Commission, 2026; Ribaudo et al., 2011). This is why simply saying that less fertilizer is always better would miss the economic problem. The outcome depends on how much nitrogen is applied and on the crop, soil, timing, placement, weather, and local hydrology.

My interest in this subject began before ECON 415. In ACE 210, I studied nitrate contamination in Central Illinois and supported a tax on excess nitrogen fertilizer together with groundwater monitoring. I mainly saw the problem as choosing a strong enough price and helping farmers adjust. Looking back, I had not answered what should be taxed, how nitrogen loss could be measured, or how much local information a uniform rule would miss. I returned to the subject because I wanted to know whether the Illinois problem sat inside one common global trend or a set of very different country experiences.

The global series and a ten-country comparison show how inorganic nitrogen fertilizer use per hectare changed from 1961 to 2023. I then use the externality model and evidence from actual policies to judge what a workable response would require. The data keep one measure at the center, while the economic analysis explains why the same application rate can create different environmental costs across places. This shifts the question from finding one correct tax rate to comparing what regulators can actually measure, monitor, and enforce.

II. Data Source

The data come from FAOSTAT and were downloaded through Our World in Data, which provides additional processing and visualization. I used two OWID downloads for different parts of the paper. The full download provides the World series and identifies 242 countries and regions. Those entries include territories, historical entities, and regional aggregates, so they are not 242 current sovereign countries. FAO’s Fertilizers by Nutrient domain covers inorganic or chemical fertilizers, including nitrogen in straight nitrogenous products and compound fertilizers (FAO, 2025; Our World in Data [OWID], 2026).

The World series contains 63 annual observations from 1961 through 2023. I use them for the global trend, descriptive statistics, and time-trend regression. I downloaded the country sample separately with OWID's "Download displayed data" option after selecting Australia, Brazil, China, France, India, Japan, Nigeria, Pakistan, South Korea, and the United States for the same years. That file contains 630 country-year observations: 10 countries with 63 years each. I used ten countries because a comparison based on only three or four cases could be driven by outliers. The group is not statistically representative of the world, but it includes different starting levels, development paths, and recent trends.

The indicator divides total agricultural use of fertilizer nutrient nitrogen by cropland area and reports the result in kg/ha (FAO, 2025). This denominator is more useful than national fertilizer totals for the question in this paper. A large country should not appear fertilizer-intensive only because it has more land. A per-capita conversion would answer a different question about nitrogen use relative to population. Here, the relevant activity is fertilizer applied to cropland, so hectares of cropland provide the clearer comparison.

The data still require caution. FAO relies mainly on its Fertilizers Questionnaire and supplements those responses with national publications and industry data. Incomplete observations may be estimated with product data or material-balance relationships, and default nutrient-content factors may be used when country-specific concentrations are unavailable (FAO, 2025). A sharp one-year change could therefore reflect reporting, estimation, political reclassification, or an actual agricultural shock. I focus on sustained movements and do not treat every one-year jump as an economic response.

This variable measures fertilizer input pressure, not environmental damage. It excludes manure and does not report crop uptake, nitrogen surplus, nitrate leaching, ammonia emissions, nitrous oxide emissions, or water quality. A country can apply less synthetic fertilizer per hectare and still have serious pollution in a livestock-intensive region. Another can increase fertilizer from a very low level and gain substantial food production without losing the same share of nitrogen. I use the series to compare long-run application patterns, not to rank environmental damage. Policy analysis still requires evidence about where the nitrogen goes and who bears the resulting costs.

III. Data Analysis

The two figures answer different questions about the trend. Figure 1 shows the global path, including its long-run rise, observed low, and peak. Figure 2 asks whether that path also describes ten individual countries. It compares five generally higher-income economies—Australia, France, Japan, South Korea, and the United States—with five large emerging or developing agricultural economies—Brazil, China, India, Nigeria, and Pakistan. A simple time-trend regression, with 1961 coded as $t = 0$, then measures the average annual change in the World series.

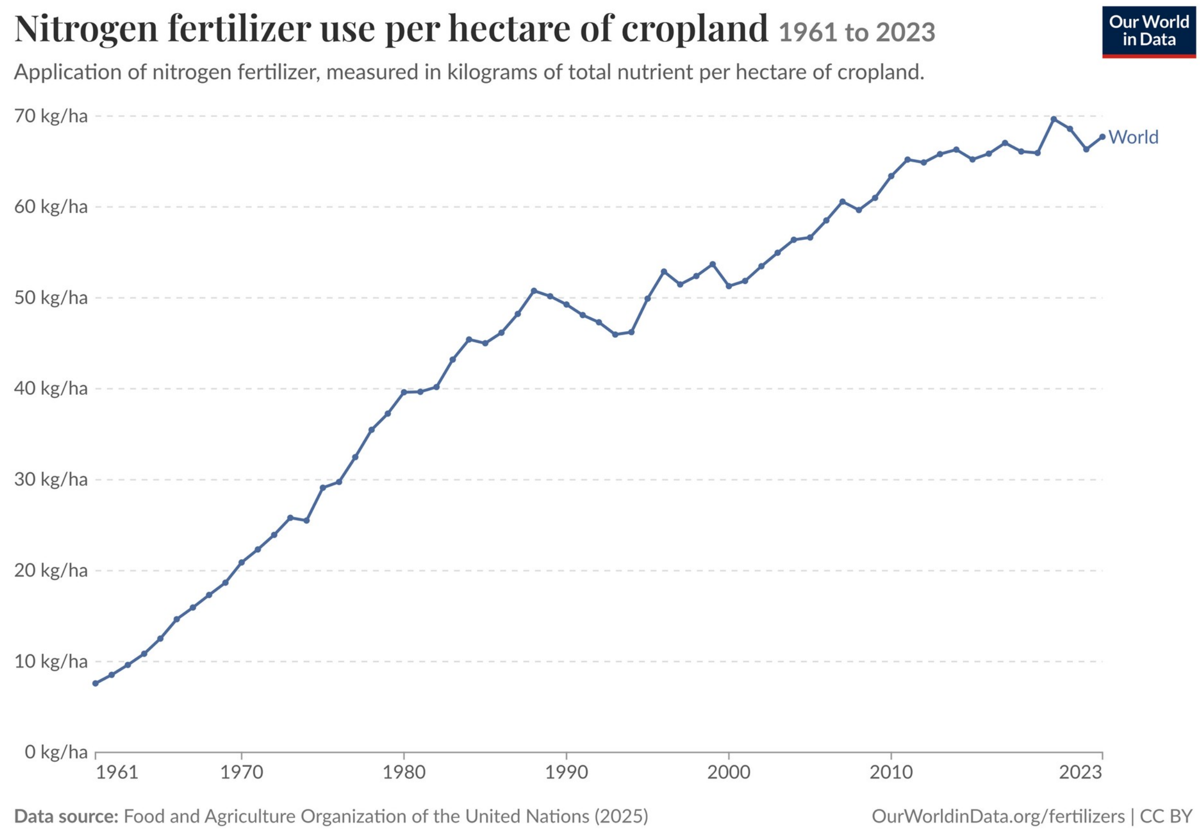

Figure 1. Global nitrogen fertilizer use per hectare of cropland, 1961–2023. Source: FAO (2025), with major processing by Our World in Data (2026).

Figure 1 shows a strong long-run increase, but the path is not a straight line. Global nitrogen fertilizer use rose from 7.59 kg N/ha in 1961, the lowest observation in the series, to 50.78 kg N/ha in 1988. It fell and moved unevenly through much of the 1990s before rising again during the 2000s. The observed peak was 69.66 kg N/ha in 2020, and the series ended at 67.72 in 2023. The 2023 value was 60.13 kg N/ha above the 1961 value, an increase of about 792 percent. That percentage looks especially large because the starting value was so low. The decline after 2020 is also too short to call a lasting reversal, since the 2023 value remained close to the peak (author's calculations from OWID, 2026).

Looking only at the first and last years would hide several important changes. Growth was rapid during the 1970s and 1980s, the early 1990s brought a sustained decline, and the period since about 2012 looks more like a high plateau. ECON 415 made me more careful about treating a date in a graph as proof of a policy effect. Fertilizer prices, crop prices, technology, weather, agricultural composition, reporting methods, and public policy can change at the same time. Figure 1 describes what happened to the variable, but it does not show what the trend would have been under a different policy.

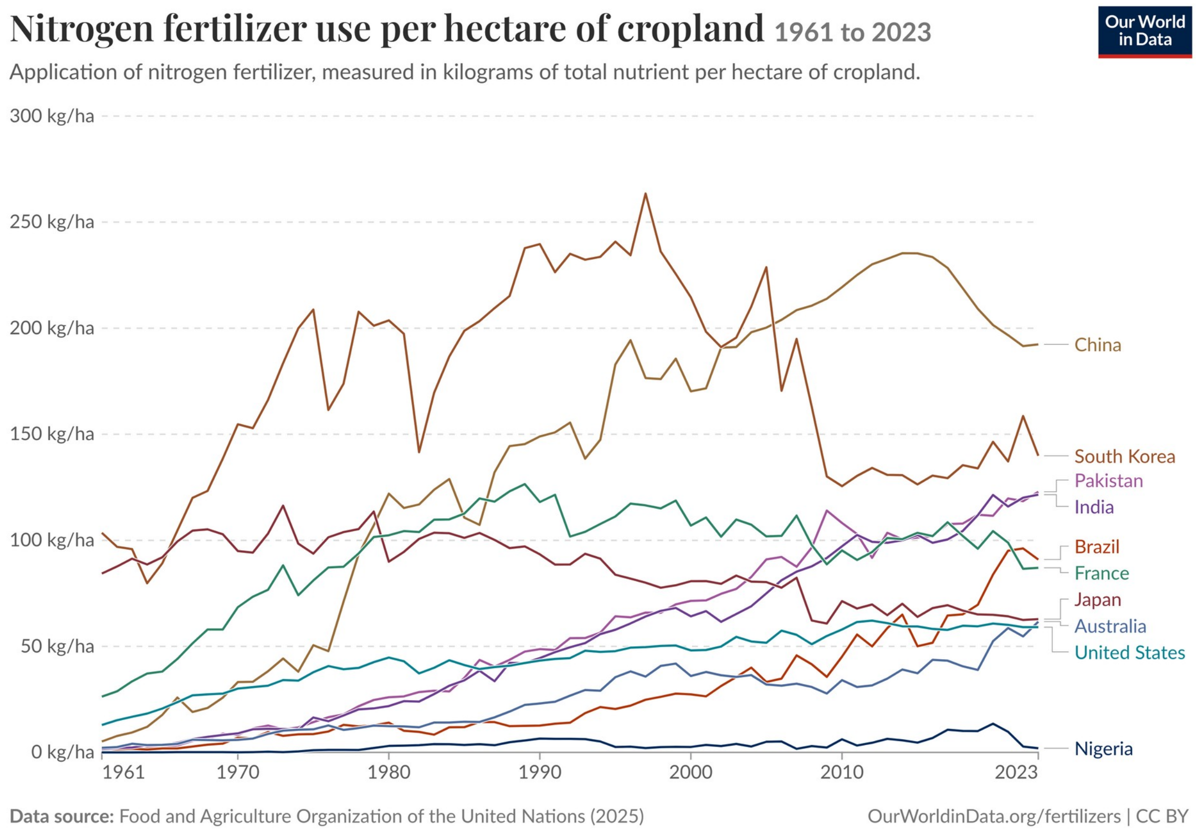

Figure 2. Nitrogen fertilizer use per hectare of cropland for a ten-country comparison, 1961–2023. Source: FAO (2025), with major processing by Our World in Data (2026).

Figure 2 shows why the global average cannot describe every country. China, India, Pakistan, and Brazil all experienced large long-run increases. China rose from 5.21 kg N/ha in 1961 to 192.39 in 2023, although its peak of 235.38 occurred in 2014. India and Pakistan started below 2 kg N/ha and reached 121.53 and 122.88 in 2023. Those endpoint values were also their sample highs. Brazil increased from 1.58 in 1961 to 90.92 in 2023, just below its 2022 peak of 96.18. These countries help explain why the global average continued to rise even after fertilizer intensity began falling elsewhere (author’s calculations from OWID, 2026).

France and Japan followed a different path. France increased from 26.27 kg N/ha in 1961 to 126.48 in 1989, then fell to 87.04 by 2023. Japan began at 84.31, reached its peak of 116.34 in 1973, and ended at 62.85. Their declines show that a country can move away from an earlier high level. The United States changed more gradually, rising from 12.96 to 59.08 and peaking at 62.14 in 2012. Australia started at only 2.17 and reached its sample high of 61.68 in 2023. Before graphing the ten countries, I expected the main difference to be their level of fertilizer use. I did not expect France and Japan to decline while Australia, India, and Pakistan were still reaching or approaching their highs. That contrast mattered more to me than the ranking in any single year. Economic development by itself does not produce one fixed fertilizer path.

South Korea and Nigeria require more caution because their series contain unusual movements. South Korea started above 100 kg N/ha, peaked at 263.51 in 1997, and ended at 139.82. Nigeria stayed close to the bottom of the common scale during most of the period, briefly reached 13.55 in 2020, and returned to 1.99 by 2023. The fertilizer series alone cannot tell me whether each jump came from prices, weather, measurement, or reporting changes. The same visible jump could reflect farm behavior or a change in reporting, so I treat the graph as a starting point rather than an explanation. I do not read those short movements as evidence that a specific policy succeeded or failed.

The common kg N/ha denominator makes these comparisons more meaningful than national totals, but it does not make farming conditions identical. A hectare of irrigated rice can have different nutrient needs and loss pathways from rain-fed grain or high-value horticulture. Figure 2 shows differences in fertilizer input intensity. It should not be read as a ranking of which country is environmentally responsible, because the figure does not measure actual nitrogen loss or the value of the crops produced.

As a simple check on the global trend, I estimated an ordinary least squares (OLS) time-trend regression. Nitrogen fertilizer use in kg N/ha is the dependent variable. I set t equal to zero in 1961, one in 1962, and 62 in 2023. The regression asks one narrow question: on average, how much did the global series change with each passing year? It contains no policy variables, prices, income measures, or country-level controls, so it cannot explain why fertilizer use changed.

Table 1. Linear time-trend regression for global nitrogen fertilizer use intensity, 1961–2023

Term	Estimate	OLS SE	t statistic	95% CI	p value
Intercept	15.018	1.119	13.426	[12.782, 17.255]	< .001
Time trend (t)	0.968	0.031	31.105	[0.906, 1.030]	< .001

Note. Dependent variable: kg N/ha of cropland. t = 0 in 1961. Ordinary least squares standard errors; $n = 63$; $R^2 = .941$. The model is descriptive, not causal. Source: Author’s calculations from OWID (2026).

The estimated time coefficient is 0.968 kg N/ha per year, with an OLS standard error of 0.031 and a 95 percent confidence interval from 0.906 to 1.030. It is statistically different from zero at conventional levels. The model has an R-squared of 0.941 across 63 observations. In simple terms, the fitted line rises by almost one kg N/ha for each year after 1961. The intercept is 15.018, which is the fitted value when $t = 0$. It is not the actual 1961 observation of 7.59.

OLS puts a number on the average increase, not on its causes. Its high R-squared mainly reflects a persistent upward trend over a long sample. The fitted line rises by 0.968 kg N/ha per year, while Figure 1 preserves the faster growth, decline, and recent stabilization that a straight line misses. I use the regression as a check against calling the series stable, not as evidence that time itself changed fertilizer use.

IV. Economic Analysis

I first use the negative production externality model to explain the problem. I then compare regulation, nitrogen charges, and more flexible incentives. Following Ribaudo et al. (2011), I ask how strongly each option changes incentives, how much land it covers, how closely it targets nitrogen loss, how much flexibility it leaves farmers, and what it costs to implement.

The problem is not that every unit of nitrogen fertilizer is harmful. A farmer receives crop value from fertilizer, but some costs from lost nitrogen fall on other people. Downstream residents can face drinking-water risks, utilities can pay more for treatment, and fisheries and recreation users can lose value. When those costs are absent from the farmer's fertilizer or output price, the private application decision can exceed the level that maximizes total social value (Ribaudo et al., 2011).

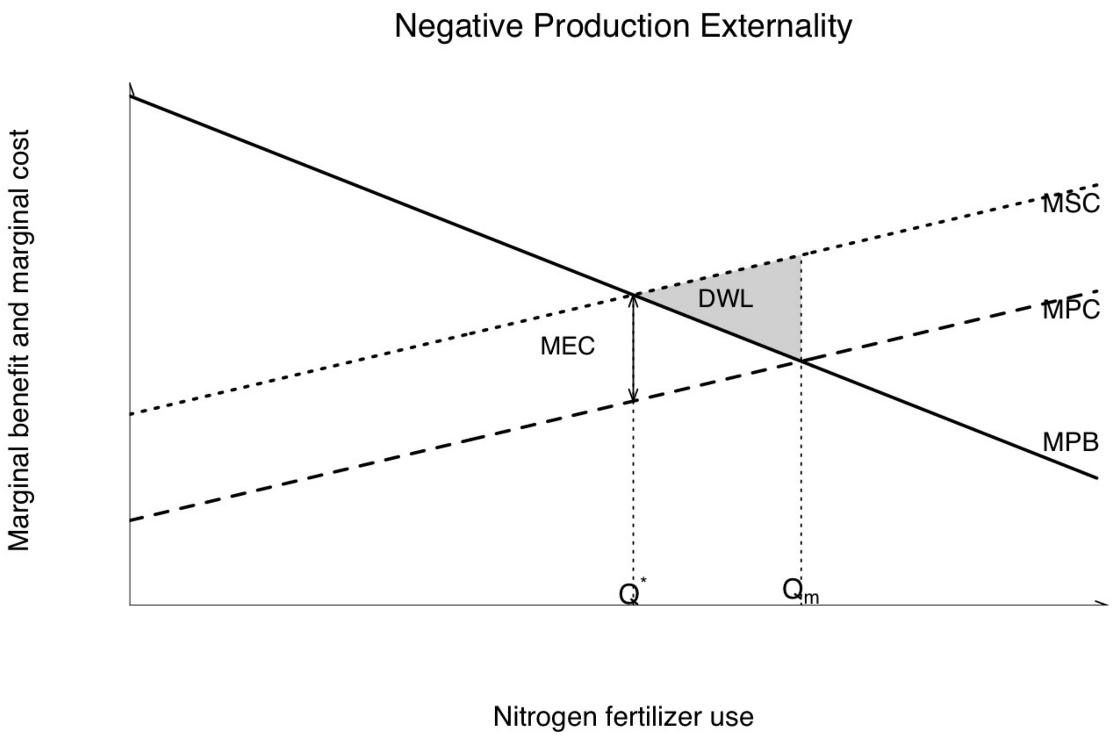

Figure 3. Negative production externality from nitrogen fertilizer use. Source: Author's illustration based on the ECON 415 model.

Figure 3 shows the externality in marginal terms. Marginal private benefit (MPB) is the crop value of the next unit of nitrogen, while marginal private cost (MPC) is the farmer's own cost. Marginal social cost (MSC) lies above MPC because it adds marginal external cost. The private market quantity is Qm, where MPB equals MPC. The efficient point is Q*, where MPB equals MSC. Units between Q* and Qm create deadweight loss because their social cost exceeds their social benefit. With perfect information, a corrective charge equal to marginal external cost at Q* would align private and social incentives (Mankiw, 2009).

Figure 1 measures fertilizer application, not marginal environmental damage. It therefore cannot identify Q* or the appropriate corrective charge. Actual nitrogen loss depends on location, soil, crop uptake, timing, and weather. The FAO measure also excludes manure and records fertilizer application instead of nitrogen surplus. A tax on all synthetic fertilizer would only approximate a Pigouvian tax on the harm. It could tax fertilizer that is taken up by crops while missing losses from manure or nitrogen already stored in the soil. This point changed how I read my ACE 210 recommendation. A fertilizer tax may still help, but taxing the input is not the same as pricing the external damage.

The property-rights problem is practical: downstream users cannot easily identify which farm caused a particular loss in water quality, and farmers do not face a market price for each unit of nitrogen that leaves their fields. Water has common-pool features because many users share it, but a farmer applying fertilizer is not extracting a common stock. The closer fit for this paper is a negative production externality. Coase (1960) asks whether bargaining could improve the result after rights are assigned. Bargaining is easier when there are few parties, damage is observable, and negotiation is inexpensive. Nutrient pollution instead involves many farms and water users, uncertain transport, time lags, weather, and difficult source attribution. Finding the parties, measuring damage, negotiating, monitoring, and enforcing an agreement make individual bargaining unrealistic for many watersheds.

The relevant comparison is between imperfect institutions. Leaving fertilizer use unregulated avoids administrative costs but leaves part of the damage unpriced. A uniform rule creates a clear floor, but it may impose very different costs across farms. Taxes and trading preserve more choice, yet they depend on measurement and enforcement. I judge each option by the agricultural and environmental value it creates after counting compliance, administration, monitoring, and political costs (Coase, 1960; Anderson & Libecap, 2014).

Hayek's knowledge problem is relevant because farmers and agronomists know details that a central regulator cannot observe continuously. They see soil conditions, crop response, machinery limits, rainfall forecasts, labor constraints, and the timing of field operations (Hayek, 1945). A uniform technology rule may require an expensive practice even when another method could reduce loss at a lower cost. A price or performance target leaves more room for local information, but the regulator still has to define a tax base, baseline, cap, or indicator that can be verified.

The first approach is command-and-control regulation. The European Union's Nitrates Directive requires monitoring, identifies nitrate-vulnerable zones, and makes action programs compulsory in those zones. It generally limits livestock-manure application to 170 kg N/ha per year and also addresses timing and crop needs (European Commission, 2021). This manure rule is not identical to the synthetic-fertilizer variable in my data, but both are parts of the larger nutrient-pollution problem. The Directive sets a clear minimum, focuses attention on higher-risk areas, and can block extreme practices even when regulators do not know a precise marginal damage price. Its costs include administration, enforcement, and the risk that one common rule imposes very different marginal abatement costs across farms.

The Directive also shows why an effective policy is not always the least-cost policy. The European Commission's July 2026 evaluation found that the Directive was effective overall and encouraged more efficient fertilizer use, although water quality can respond slowly and implementation remains uneven (European Commission, 2026). The Commission estimated annual implementation costs of about EUR 2.8–3.1 billion and benefits of about EUR 10–22 billion, roughly three to seven times the costs. It also estimated the wider social cost of agricultural nitrogen pollution in Europe at EUR 68–182 billion per year. These estimates apply to the European Union, not the world. The benefit estimate makes me less willing to dismiss standards as inefficient by definition. The declines in France and Japan show that fertilizer intensity can fall after reaching a high level, although Figure 2 cannot attribute those declines to the Directive. I see the Directive as a useful environmental floor, not as proof that one uniform rule is the least-cost answer everywhere.

A second approach uses incentive-based taxes or charges to change the price of nitrogen use. One option is a charge on nitrogen inputs or, if measurement is possible, on farm nitrogen surplus. In the textbook Pigouvian case, a charge equal to marginal external cost at Q* gives farms a reason to reduce nitrogen until their marginal abatement cost equals the charge. Farms with less expensive options reduce more, while farms facing higher costs reduce less. This can reach a common reduction target at a lower total resource cost (Mankiw, 2009). Revenue could also support monitoring or agronomic assistance. The practical problem is choosing the right charge. Damage is uncertain, fertilizer purchases are an imperfect proxy for loss, and returning revenue does not make the policy costless.

Evidence from Switzerland makes me cautious about relying only on an input tax. Schmidt et al. (2017) found low responsiveness to nitrogen-input charges in their agricultural model. Even a levy several times the nitrogen price produced a relatively modest modeled reduction in nitrogen surplus. I would not use one Swiss model to predict every country, but its result still matters: nitrogen demand may respond slowly unless the charge is large. A surplus charge would better target the gap between nitrogen entering and leaving a farm, but it would require nutrient accounts and rules for manure, biological fixation, and changes in soil nitrogen. Better targeting brings higher administrative costs. I would introduce a verified surplus charge gradually instead of treating every fertilizer purchase as equal environmental damage.

A third group of policies lowers adoption risk or rewards verified reductions. Cost-sharing for soil testing, split application, cover crops, precision equipment, or nutrient plans can make a change easier and may face less political resistance than a tax. The weakness is that public money can go to farms that would have adopted anyway, while voluntary programs may miss acres with the largest losses (Ribaudo et al., 2011). An Illinois modeling study proposed nitrogen insurance that rewards reductions above recommended application rates while limiting yield risk. Across simulated maize fields, it reduced modeled nitrate leaching and created gains for eligible farmers and insurers (Mandrini et al., 2026). It was a model rather than a completed national program, but I find the approach useful because it treats yield uncertainty as part of the incentive problem.

The same group also includes tradable water-quality credits, which use differences in marginal abatement costs rather than directly paying for a specific practice. A farm or regulated point source with a low-cost reduction can produce verified credits and sell them to a source facing higher costs. The U.S. Environmental Protection Agency permits trading frameworks for nitrogen and phosphorus when they remain consistent with Clean Water Act water-quality standards (U.S. Environmental Protection Agency [EPA], 2026a). In practice, nonpoint reductions are difficult to measure, baselines can be disputed, and rainfall changes the outcome. Thin markets and badly designed trades can also create local pollution hotspots (EPA, 2026b; Ribaudo et al., 2011). I would use trading as a compliance option under a water-quality floor, not as a substitute for that floor. Otherwise, a low-cost credit in one place could leave too much damage in another.

Yandle changed the question I ask about regulation. Clean-water organizations and drinking-water users can support stronger rules for public reasons. Fertilizer firms, farm organizations, technology vendors, or established farms may support the same proposal because it offers subsidies, exemptions, or higher barriers to entry (Yandle, 1999). That does not prove the policy is inefficient, but it makes the distribution of permits, payments, and compliance costs part of the analysis. I also need to ask whether a rule blocks entry or innovation.

My preferred policy would not rely on a fertilizer tax alone. I would begin with enforceable water-quality limits and minimum nutrient-management rules in vulnerable areas. Farms above a practical size threshold could keep consistent nutrient accounts, allowing a gradually introduced charge on verified surplus. Flexible compliance and carefully checked credits could then shift reductions toward lower-cost sources. Some revenue should pay for independent monitoring and temporary help with measurement or low-loss technology, especially where fixed compliance costs are high for smaller farms. This approach is more complicated than a single tax, but it combines a clear environmental floor with incentives to find less expensive reductions.

V. Conclusion

Global fertilizer intensity rose sharply over the full sample, but the country comparison shows that the increase was not universal. The World series rose from 7.59 kg N/ha in 1961 to 67.72 in 2023 and peaked at 69.66 in 2020. The regression estimates an average increase of 0.968 kg N/ha per year. France and Japan moved down after earlier peaks, China declined from its 2014 peak while remaining high, and India and Pakistan reached their sample highs in 2023. The global average shows the broad direction, but it is not enough to design country-level policy. The policy implication is less simple: a response must address the externality without ignoring measurement and compliance costs, and any prediction after 2023 has to remain cautious.

The main economic problem is a negative production externality. Farmers consider crop value and the private cost of fertilizer, but they may not face all downstream and atmospheric damage. The data cannot identify Qm or Q* because fertilizer application is not the same as nitrogen loss. Property-rights bargaining is difficult when many farms and water users face uncertain transport and high transaction costs. Hayek's knowledge problem supports flexible rules, while differences in marginal abatement costs support shifting reductions toward lower-cost sources. Public-choice theory also made me ask who gains from the final policy. A regulation may improve water quality while still giving subsidies, exemptions, or entry advantages to well-organized groups.

My preferred policy has changed since ACE 210. I previously favored a stand-alone tax on excess fertilizer. I now think the first difficulty is deciding what "excess" means and measuring it consistently. I would combine water-quality and nutrient-management minimums in vulnerable places with nutrient accounts for larger farms and a gradual charge on verified surplus. Trading would be allowed only where baselines and local safeguards can be enforced. Revenue would support independent monitoring and temporary insurance or measurement assistance. Better targeting and lower total compliance costs are the main benefits. Administrative expense, privacy concerns, measurement error, and burdens on small farms are the main tradeoffs. I accept the added administrative cost only because this approach targets surplus more closely than a tax on every fertilizer purchase.

My ACE 210 paper made me see nitrogen pollution mainly as a local question of fertilizer rates and groundwater monitoring. ECON 415 pushed me to ask who can measure the losses, who pays for compliance, and how organized groups may shape the final rule. I still see a price incentive as useful, but I no longer think that selecting a tax rate completes the analysis. The EU evaluation also changed my earlier view that command-and-control should mainly serve as an inefficient benchmark. A rule can produce benefits well above its implementation costs and still need more flexible execution (European Commission, 2026).

Based on the opposing paths in Figure 2, I expect global nitrogen fertilizer intensity to remain near a high plateau or grow more slowly after 2023. Food demand and recent increases in countries such as India and Pakistan may push the average upward. Efficiency gains and lower intensity in some high-use economies may push it downward. Fertilizer prices, weather, conflict, and reporting changes could continue to create annual volatility. The observed series ends in 2023, so this is a prediction for later years and not an additional data observation.

VI. References

The references include the fertilizer data, the course readings, and the policy evidence used in the analysis.

Anderson, T. L., & Libecap, G. D. (2014). *Environmental markets: A property rights approach*. Cambridge University Press.

Coase, R. H. (1960). The problem of social cost. *Journal of Law and Economics, 3*, 1–44. https://doi.org/10.1086/466560

European Commission. (2021). *Report from the Commission to the Council and the European Parliament on the implementation of Council Directive 91/676/EEC concerning the protection of waters against pollution caused by nitrates from agricultural sources based on Member State reports for the period 2016–2019* (COM(2021) 1000 final). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021DC1000

European Commission. (2026). *Executive summary of the evaluation: Commission staff working document evaluation of the Nitrates Directive (Council Directive 91/676/EEC)* (SWD(2026) 235 final). https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52026SC0235

Food and Agriculture Organization of the United Nations. (2025). *FAOSTAT domain: Fertilizers by nutrient—Methodological note, release July 2025*. https://files-faostat.fao.org/production/RFN/RFN_EN_README.pdf

Hayek, F. A. (1945). The use of knowledge in society. *American Economic Review, 35*(4), 519–530.

Mandrini, G., Davidson, E. A., Nafziger, E. D., Eagle, A. J., Pittelkow, C. M., Gentry, L. F., Schnitkey, G. D., Guan, K., Martin, N. F., Chen, Z., Alles, K., & Ciampitti, I. A. (2026). A market-based insurance approach aligns environmental and economic outcomes in maize nitrogen management. *Communications Earth & Environment, 7*, Article 26. https://doi.org/10.1038/s43247-025-03008-3

Mankiw, N. G. (2009). Smart taxes: An open invitation to join the Pigou Club. *Eastern Economic Journal, 35*(1), 14–23. https://doi.org/10.1057/eej.2008.43

Our World in Data. (2026). *Nutrient nitrogen n (total)—Use per area of cropland (kilograms per hectare)—UN FAO* [Data set; original data from the Food and Agriculture Organization of the United Nations, 2025, with major processing by Our World in Data]. Retrieved August 9, 2026, from https://ourworldindata.org/grapher/nitrogen-fertilizer-application-per-hectare-of-cropland

Ribaudo, M., Delgado, J., Hansen, L., Livingston, M., Mosheim, R., & Williamson, J. (2011). *Nitrogen in agricultural systems: Implications for conservation policy* (Economic Research Report No. 127). U.S. Department of Agriculture, Economic Research Service. https://ers.usda.gov/publications/44919

Schmidt, A., Necpalova, M., Zimmermann, A., Mann, S., Six, J., & Mack, G. (2017). Direct and indirect economic incentives to mitigate nitrogen surpluses: A sensitivity analysis. *Journal of Artificial Societies and Social Simulation, 20*(4), Article 7. https://doi.org/10.18564/jasss.3477

U.S. Environmental Protection Agency. (2026a). *Water quality trading policy*. https://www.epa.gov/npdes/water-quality-trading-policy

U.S. Environmental Protection Agency. (2026b). *Frequent questions about water quality trading*. https://www.epa.gov/npdes/frequent-questions-about-water-quality-trading

Yandle, B. (1999). Bootleggers and Baptists in retrospect. *Regulation, 22*(3), 5–7. https://www.cato.org/sites/cato.org/files/serials/files/regulation/1999/10/bootleggers.pdf